OralGuard: A Smartphone-Accessible Platform for Oral Cancer Risk Awareness Using Content-Based Image Retrieval

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Abstract

Oral cancer carries a disproportionately high mortality burden relative to its visual accessibility, largely because most cases are identified only at advanced stages. This paper describes OralGuard, a smartphone-accessible platform designed to raise awareness of oral cancer risk through guided self-examination, behavioral risk assessment, and visual case comparison. The core technical component is a content-based image retrieval (CBIR) system that compares a user-submitted oral lesion image against a database of clinically annotated cases, returning visually similar benign and malignant reference images alongside similarity scores. Rather than producing a diagnostic label, the system presents comparative visual evidence intended to help users recognize potentially concerning presentations and seek timely professional consultation. Feature extraction is performed using a MobileNetV2 backbone, with retrieval based on cosine similarity over L2-normalized embeddings. The platform is accessible via a Flutter mobile application and a web interface, with inference handled by a Python FastAPI backend. A structured behavioral risk questionnaire capturing tobacco use, alcohol consumption, and other established risk factors complements the image matching component. The system is evaluated as a proof-of-concept awareness tool, with acknowledged limitations regarding dataset scale and the absence of formal user validation.

Keywords: Oral cancer awareness, Content-based image retrieval, Deep learning, Smartphone screening, MobileNetV2, FastAPI, Self-examination

1 Introduction

Oral cancer accounts for roughly 5% of all cancer cases globally, but in India the burden is considerably higher, representing approximately 40% of diagnosed cancers, with nearly 60,000 new cases annually [1]. The survival gap between early and late-stage diagnosis is stark: five-year survival rates at Stage I approach 84%, while Stage IV presentations fall below 20%. Yet the majority of cases are still identified late [2]. This epidemiological pattern persists despite oral mucosa being directly accessible to visual inspection, which points to a failure not of clinical capability but of awareness and access. Many individuals do not recognize early warning signs, do not have ready access to oral health specialists, or simply do not seek evaluation until symptoms become pronounced.

Artificial intelligence-based tools have attracted substantial research interest as a potential bridge between community-level risk exposure and specialist-level evaluation. Mobileintegrated classification models can in principle analyze intraoral images at the point of capture, providing rapid preliminary assessment without requiring specialist attendance [2]. In practice, however, classification-based AI systems present real limitations for community deployment. Diagnostic performance varies substantially across datasets and imaging conditions [3], and the outputs—typically probability scores for benign versus malignant categories—are poorly suited to non-clinical users who lack the context to interpret them meaningfully.

Content-based image retrieval offers a complementary and arguably more suitable paradigm for awareness-oriented applications. Rather than assigning a probability, a CBIR system retrieves visually similar cases from a reference database and presents them alongside known clinical outcomes [4]. The user can see how a submitted image compares to confirmed cases, forming an intuitive visual judgment that may be more actionable than an abstract score. This approach has demonstrated practical utility in dermatology [5] and radiology [6], but its application to oral cancer awareness tools has not been systematically explored.

OralGuard is a platform designed around this retrieval paradigm. It integrates guided oral self-examination, a structured behavioral risk questionnaire, and CBIR-based image comparison into a single accessible interface, available as both a Flutter mobile application and a Gradio web application. The platform does not produce clinical diagnoses. Its purpose is to raise awareness, support self-monitoring, and lower the threshold for seeking professional evaluation, particularly in settings where access to oral health specialists is limited.

2 Literature Review

2.1 Content-Based Medical Image Retrieval

Content-based medical image retrieval refers to computational approaches that locate images from large repositories by analyzing intrinsic visual characteristics, including texture, spatial structure, color distribution, and morphological patterns, rather than relying on textual metadata. Early CBMIR systems established the foundational architecture of feature extraction, indexing, and distance-based similarity matching using handcrafted descriptors [4, 6]. These principles remain relevant in contemporary systems, though the feature extraction component has been substantially transformed by deep learning.

Convolutional neural networks extract high-dimensional semantic representations from raw images with considerably greater fidelity than manually engineered descriptors, particularly for capturing subtle morphological variation across medical image classes [7, 8]. Modern CBMIR systems typically pair deep feature encoders with scalable nearest-neighbor search strategies to enable practical retrieval from large-scale repositories [9]. Applications span radiology, where retrieved cases support differential diagnosis and treatment planning [6]; digital pathology, where comparison of histopathological patterns aids diagnostic consistency [10]; and dermatology, where systems comparing lesion images against annotated dermoscopic databases have reached clinically relevant performance levels [5, 11].

In oral oncology, the predominant AI research direction has been supervised classification rather than retrieval [13, 14]. Classification frameworks generate decision boundaries and probability outputs but offer limited interpretive support, particularly for lay users. Retrieval-based approaches can partially address this interpretability gap by grounding outputs in visually verified clinical precedents. The application of CBMIR within an oral cancer awareness platform specifically designed for community use has not been previously reported in the literature.

2.2 Mobile and Point-of-Care Screening

The convergence of high-resolution imaging, connectivity, and computational capability in consumer smartphones has made point-of-care disease screening increasingly feasible, particularly in low- and middle-income settings where specialist access is constrained. A large-scale study conducted in India demonstrated that community health workers using mobile devices for intraoral imaging, combined with teleconsultation and specialist review, could effectively triage high-risk lesions and reduce diagnostic delays [15]. This establishes that structured mobile screening workflows, even those requiring human oversight, can function meaningfully at the community level.

In dermatology, deep learning classifiers trained on skin lesion images have matched or exceeded dermatologist-level performance in controlled evaluation settings [5], and human-AI collaborative evaluation appears to outperform either alone [11]. Smartphone-based deep learning systems for melanoma have demonstrated comparable results [16]. The evidence from these adjacent domains is broadly supportive of the potential for mobile AI-assisted tools to enhance early identification of visually detectable cancers, while not being directly transferable given differences in imaging conditions and dataset availability.

For an awareness-oriented application, the primary value of mobile accessibility is reach rather than clinical precision. A platform that can be used by individuals in community settings to compare observed lesions against reference cases and complete a structured risk assessment represents a supplementary pathway that may prompt earlier professional consultation. It functions as a triage and awareness instrument rather than a diagnostic one, and should be evaluated in that context.

3 Methodology

3.1 System Architecture

OralGuard is organized as a three-tier architecture. The presentation layer consists of a Flutter-based mobile application and a Gradio web interface, both providing access to the platform's three core modules: the Image Matcher, the Risk Screener, and the Self-Exam Guide. The backend layer is a Python FastAPI server handling image preprocessing, feature extraction, similarity computation, and response formatting. Communication between layers is handled via RESTful API calls, preserving modularity and simplifying future extension.

All inference occurs server-side. The mobile application and web interface function as capture and display layers, with no heavy computation performed on the user's device. This design makes the platform accessible on low-end smartphones with stable internet connectivity. Privacy is addressed architecturally: no personal data is collected, uploaded images are processed transiently and not stored or logged, and no identifying information is required or retained after a session ends.

3.2 Dataset

3.2.1 Dataset Description

The CBIR index and feature extraction model were built using the Oral Cancer Image Dataset published on Kaggle by Zaid Py [18]. The dataset comprises oral lesion images categorized into two classes: benign and malignant. It was selected for its direct relevance to intraoral lesion presentation and its availability for research use. The full dataset was used to construct the reference embedding index, against which query images are matched at inference time.

It is worth noting that the dataset is relatively limited in scale compared to what would be required for a clinically deployed system. This is acknowledged as a significant constraint and is discussed further in Section 4.

3.2.2 Preprocessing Pipeline

All images were resized to 224×224 pixels to match MobileNetV2 input requirements. Pixel values were normalized using the ImageNet distribution statistics (mean [0.485, 0.456, 0.406], standard deviation [0.229, 0.224, 0.225]) to align preprocessing with the pretrained model's training conditions. Data augmentation was applied during training, including horizontal flipping and random brightness variation. These strategies were selected for their clinical plausibility, as mirrored lesion appearances and variable illumination conditions are consistent with real-world intraoral imaging.

3.3 Feature Extraction

3.3.1 Backbone Architecture

The feature extraction backbone is MobileNetV2 [17], pretrained on ImageNet. It was selected primarily for deployment efficiency: the depthwise separable convolution structure substantially reduces parameter count and inference latency relative to full convolution architectures, a meaningful consideration for a server expected to handle concurrent requests from a web-accessible endpoint.

The classification head of the pretrained network was removed. The modified architecture consists of the convolutional feature extraction layers, followed by adaptive average pooling to a 1×1 spatial output, and flattening to produce a 1280-dimensional embedding vector. Each embedding is subsequently L2-normalized to unit length, enabling cosine similarity to be computed as a dot product.

3.3.2 Implementation Details

The model is instantiated using PyTorch's `torchvision` library with ImageNet pretrained weights. Inference runs in evaluation mode with gradient computation disabled. The extraction pipeline is as follows:

```
backbone = models.mobilenet_v2(weights=MobileNet_V2_Weights.IMAGENET1K_V1)
```

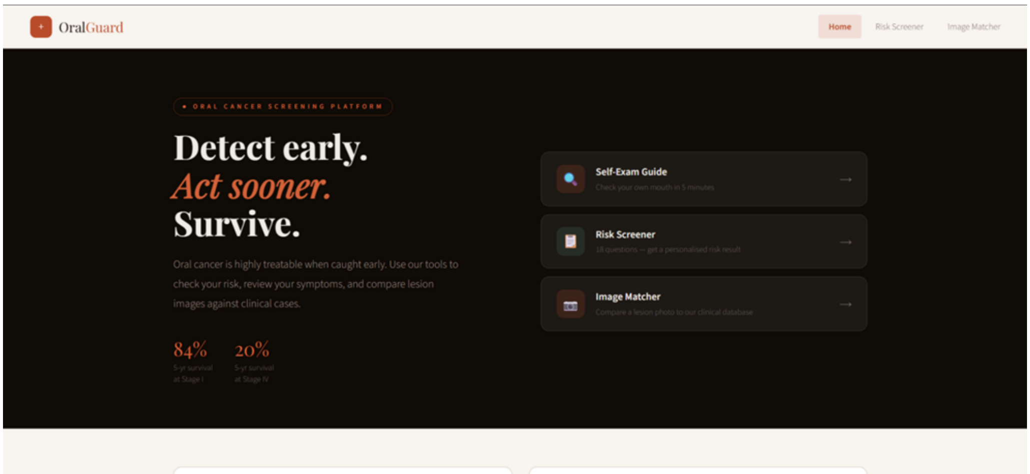

Fig. 1a. OralGuard web application home screen

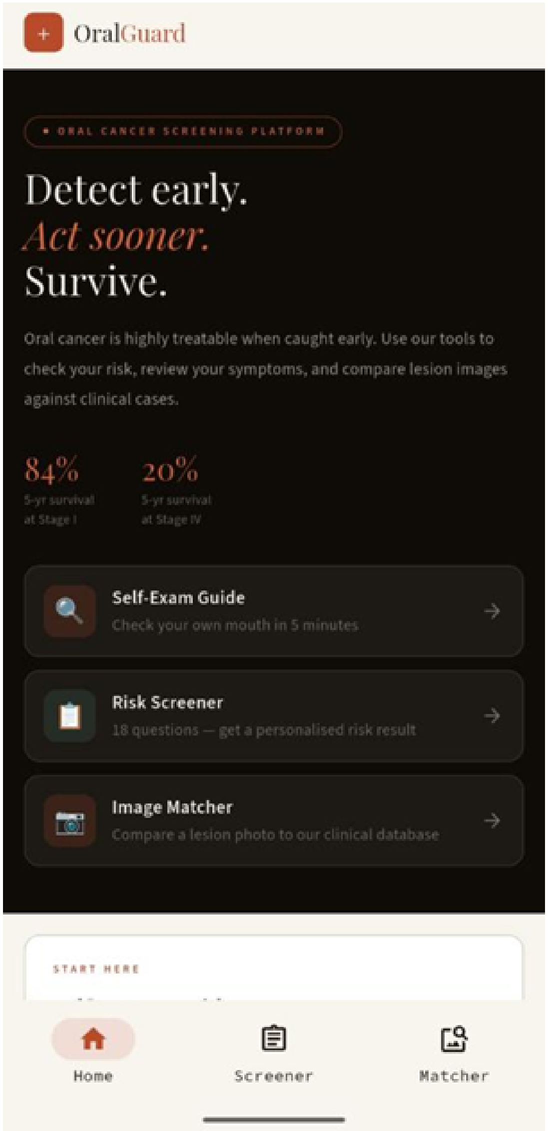

Fig. 1b. OralGuard mobile application home screen

```
model = nn.Sequential(
    backbone.features,
    nn.AdaptiveAvgPool2d((1, 1)),
    nn.Flatten()
)
embedding = model(tensor).squeeze().cpu().numpy()
embedding = embedding / (np.linalg.norm(embedding) + 1e-8)
```

The epsilon term (`1e-8`) in the denominator prevents numerical instability for near-zero vectors, which may occur with low-quality or unusual input images.

3.4 Content-Based Image Retrieval Pipeline

3.4.1 Reference Database Construction

All images in the dataset were passed through the feature extraction pipeline, and the resulting embeddings were stored in a compressed NumPy archive (`cbir_index.npz`) alongside their corresponding file paths and clinical labels. This index is loaded into memory at server startup, enabling retrieval without repeated disk access during inference.

3.4.2 Query Processing and Similarity Ranking

When a user submits an intraoral image, the backend decodes it as an RGB PIL image and passes it through the extraction pipeline. The resulting query embedding is compared against all indexed embeddings using cosine similarity. Results are separated into benign and malignant pools based on clinical labels and ranked independently by similarity score. The top six matches from each pool are returned, along with similarity scores and publicly accessible image URLs.

The decision to return separate ranked pools rather than a merged list reflects a deliberate design choice. A single merged ranking can be dominated by one category when the query strongly resembles cases of that type, which tends to anchor user interpretation prematurely. Presenting both pools in parallel preserves comparative context and is less likely to produce false reassurance or unwarranted concern, either of which could undermine the platform's awareness objective.

3.5 Risk Questionnaire

In parallel with image matching, users complete a structured questionnaire capturing established behavioral and demographic risk factors for oral cancer. Variables assessed include tobacco use (frequency and type), smokeless tobacco and betel nut consumption, alcohol use, HPV history, UV exposure, family history of oral cancer, and self-reported oral symptoms. Each response is assigned a weighted score derived from epidemiological risk association literature, and the aggregate score produces a risk classification output.

The questionnaire component serves two related purposes. It captures risk information that images cannot convey: behavioral exposures that substantially modulate oral cancer risk are not visually detectable from a lesion photograph alone. It also engages the user

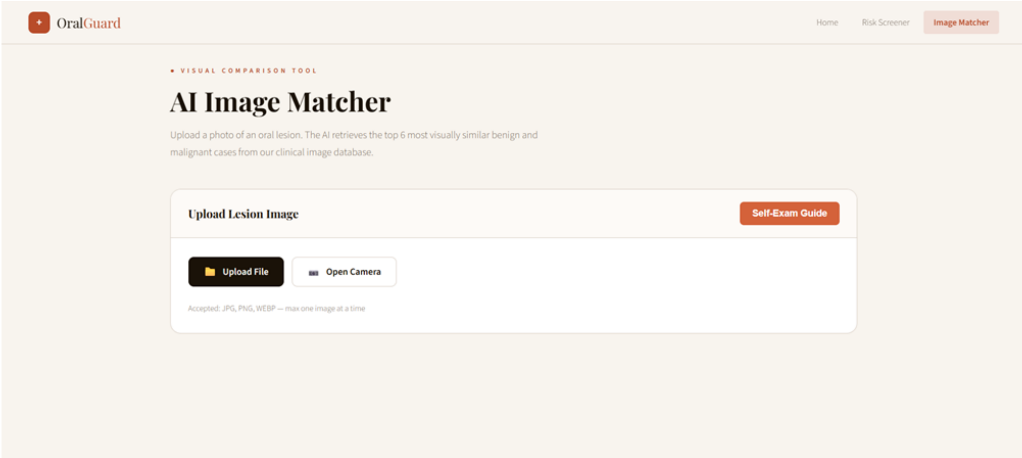

Fig. 2a. Image Matcher upload interface (web)

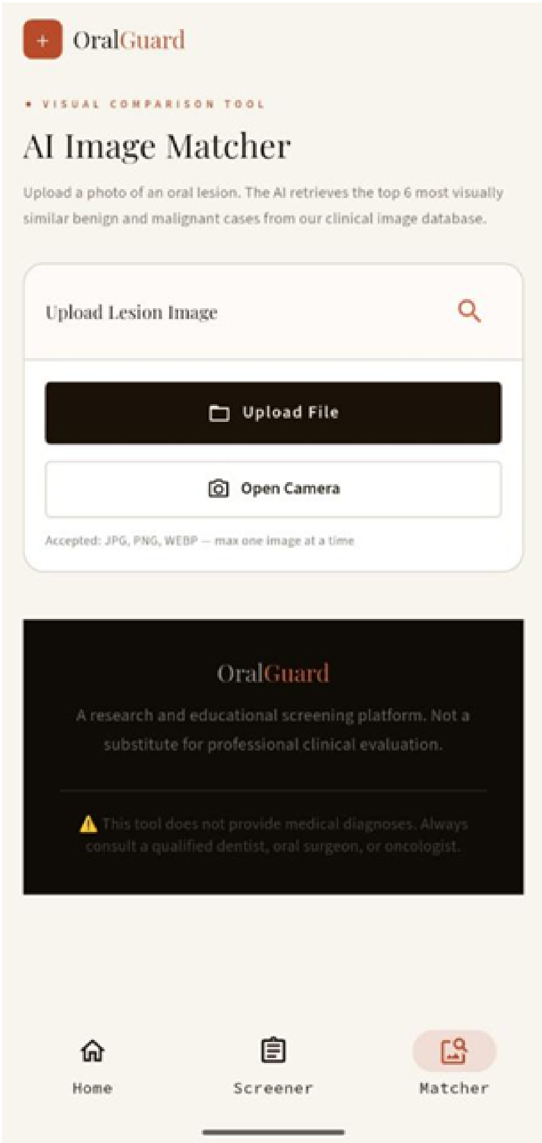

Fig. 2b. Image Matcher upload interface (mobile)

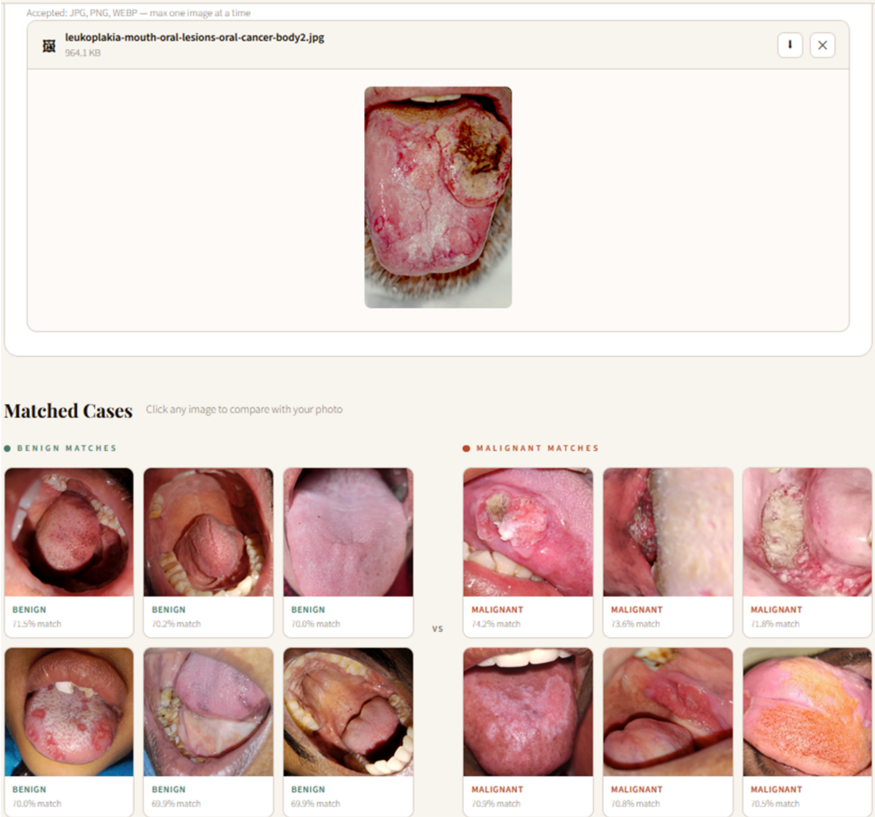

Fig. 3. CBIR matched results output showing top-6 benign and malignant reference cases with similarity scores

in a structured reflection on their own risk profile, which may independently increase awareness of modifiable risk factors regardless of the image retrieval result.

3.6 Backend API and Deployment

3.6.1 FastAPI Endpoint Design

The backend is implemented using FastAPI. The primary inference endpoint at `POST /search` accepts multipart image uploads and returns two ranked result lists. The application includes CORS middleware configured to accept requests from all origins, enabling access from both the mobile application and the web interface. A root `GET` endpoint at `/` returns a health check response.

The complete inference sequence at `/search` is: read the uploaded file into memory, decode as an RGB PIL image, extract a normalized embedding via MobileNetV2, compute cosine similarity against all indexed embeddings, separate results by label, sort each pool independently, and return the top six from each as a JSON response including similarity scores, labels, and image URLs.

3.6.2 Deployment

The FastAPI backend is containerized and deployed on Hugging Face Spaces, providing a publicly accessible inference endpoint without dedicated server infrastructure. The pretrained model and CBIR index are loaded once at container startup and held in

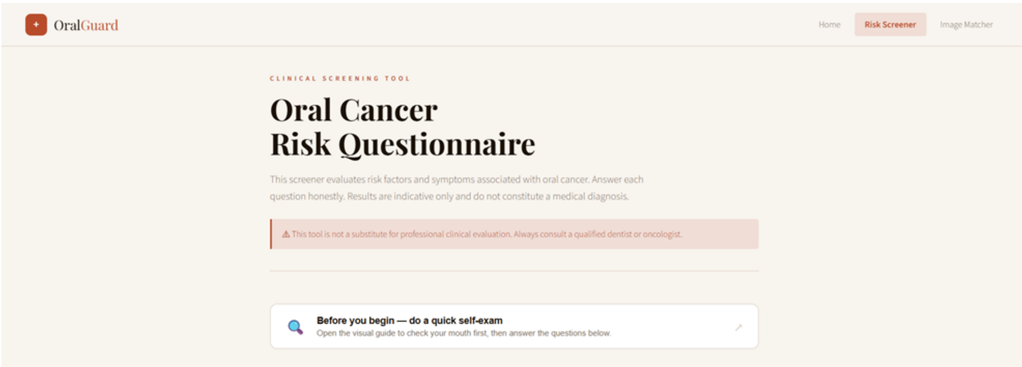

Fig. 4a. Risk Screener landing page (web)

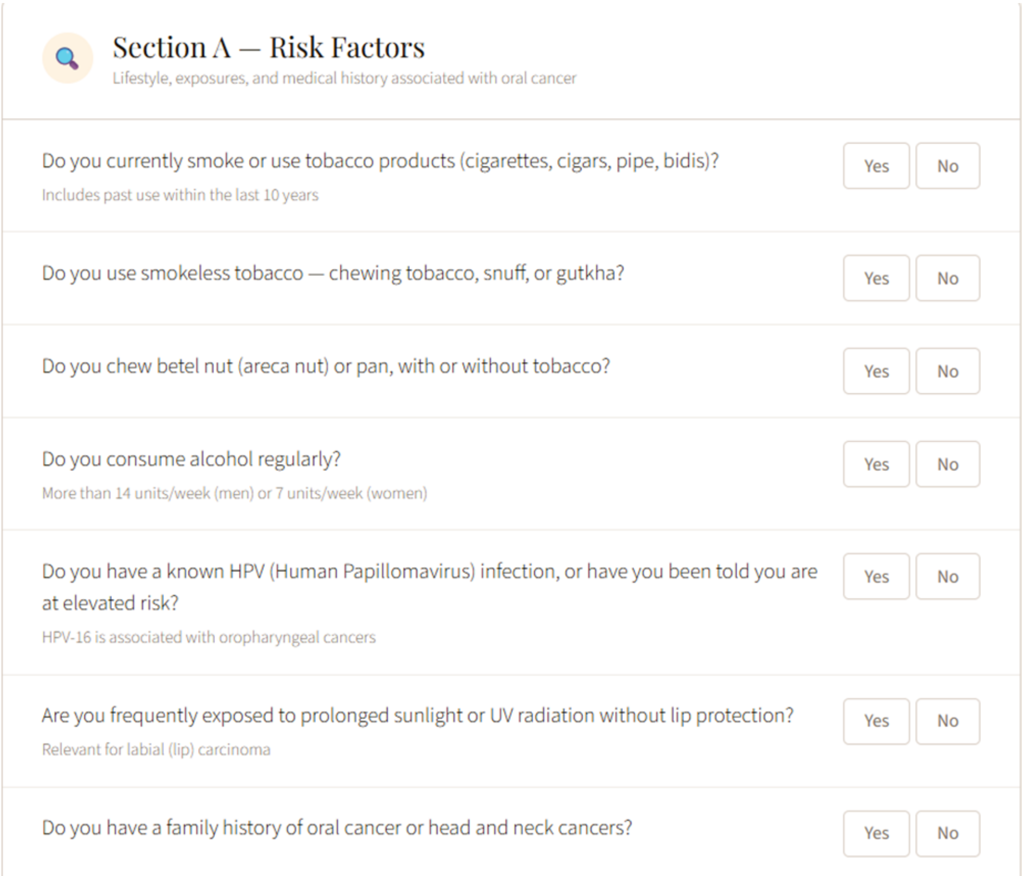

Fig. 4b. Risk factor questionnaire section (web)

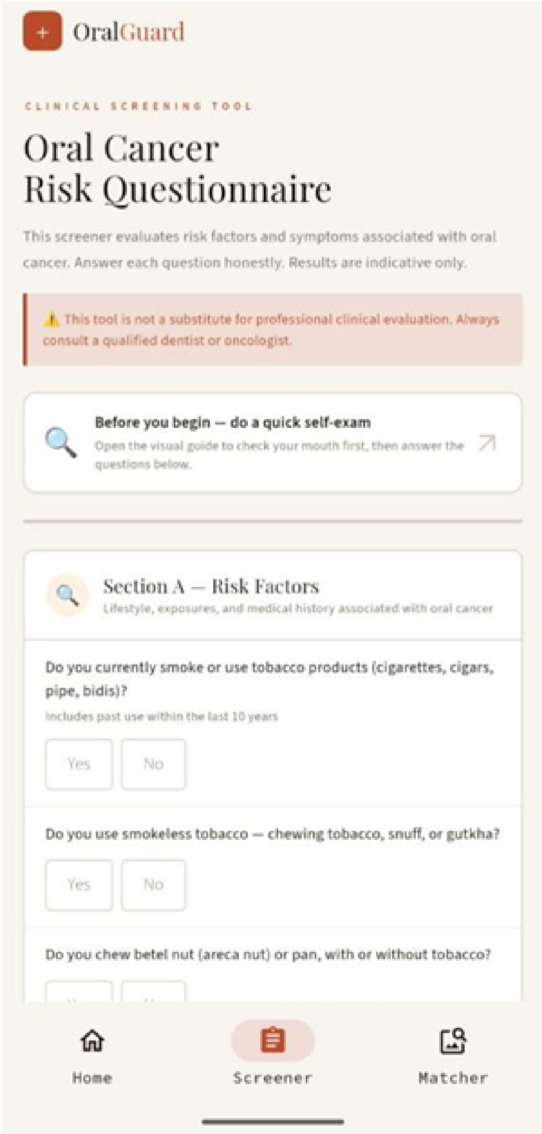

Fig. 5. Risk Questionnaire on the OralGuard mobile application

memory. The Gradio web interface and the Flutter mobile application both connect to the same backend endpoint, ensuring consistent retrieval behavior across access modalities.

3.7 User Interface and User Experience

All interfaces are designed around an accessibility-first philosophy, prioritizing minimal interaction overhead for users without clinical background. The complete workflow for a first-time user involves three steps: completing the optional self-exam guide to position and capture the image correctly, uploading the image in the Image Matcher, and optionally completing the Risk Screener. No account creation or personal data submission is required at any point.

The Self-Exam Guide provides structured anatomical instructions for intraoral self-examination, covering six oral regions: lips and corners, inner cheeks and gums, tongue, floor of mouth, palate, and throat and neck. An annotated clinical diagram illustrates each examination zone. This guided capture protocol is intended to reduce the variability inherent in user-generated intraoral images, which can otherwise vary substantially in framing, lighting, and anatomical coverage.

Retrieved reference images are presented in a side-by-side grid layout with similarity scores and clinical labels. The intent is for users to form an intuitive visual judgment by comparing their submitted image against confirmed cases, rather than relying solely on a numerical score. This design reflects the recognition that non-clinical users are generally poorly positioned to interpret probability or similarity scores in medically meaningful ways; visual comparison may be more accessible and more likely to influence behavior.

4 Results and Discussion

The platform was tested informally using held-out oral lesion images not included in the reference index. Retrieval outputs were qualitatively consistent for visually distinct presentations: queries containing erythematous patches, ulcerated lesions, or leucoplakic areas returned reference cases with recognizable visual correspondence. Queries with low-contrast or ambiguous lesion morphology produced lower peak similarity scores and less clearly differentiated result sets across both pools, which is an expected behavior given the feature extraction approach.

A recurring interpretive concern involves the relationship between cosine similarity scores and clinical risk. A high similarity score between a query image and a malignant reference case indicates visual resemblance to that case, not likelihood of malignancy. This distinction is clinically important and must be communicated clearly through interface design. The current implementation includes contextual disclaimers throughout the interface, but whether these are sufficient to prevent misinterpretation is an empirical question that requires user study validation.

The dual-pool result presentation appears to support more balanced interpretation than a single merged ranking would. When the query image strongly resembles cases from one category, a merged ranking can create a misleading visual impression of certainty. Separate pools preserve the comparative context and make it easier for users to see that visual similarity to one category does not preclude similarity to the other.

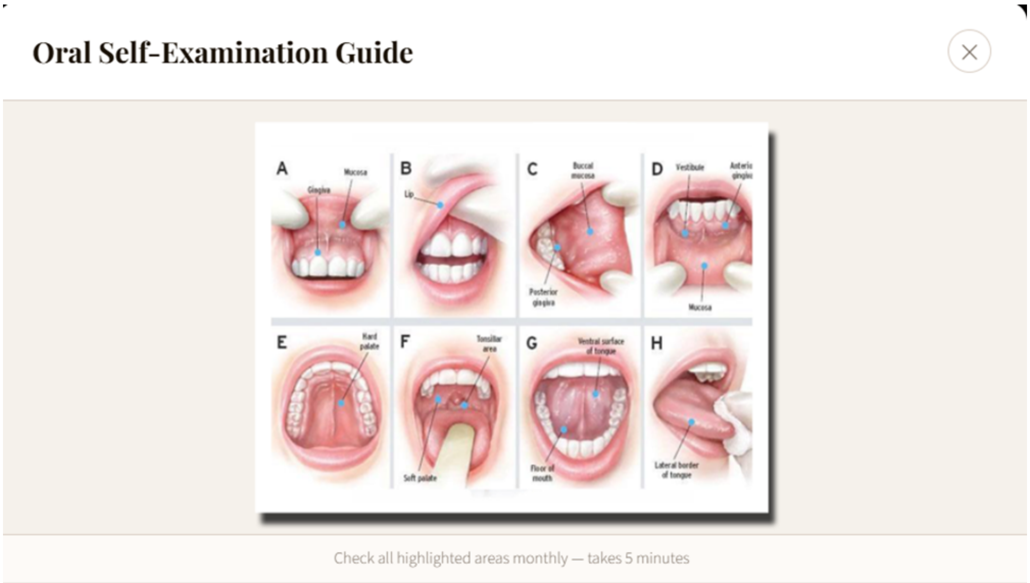

Fig. 6a. Oral self-examination anatomical guide (web)

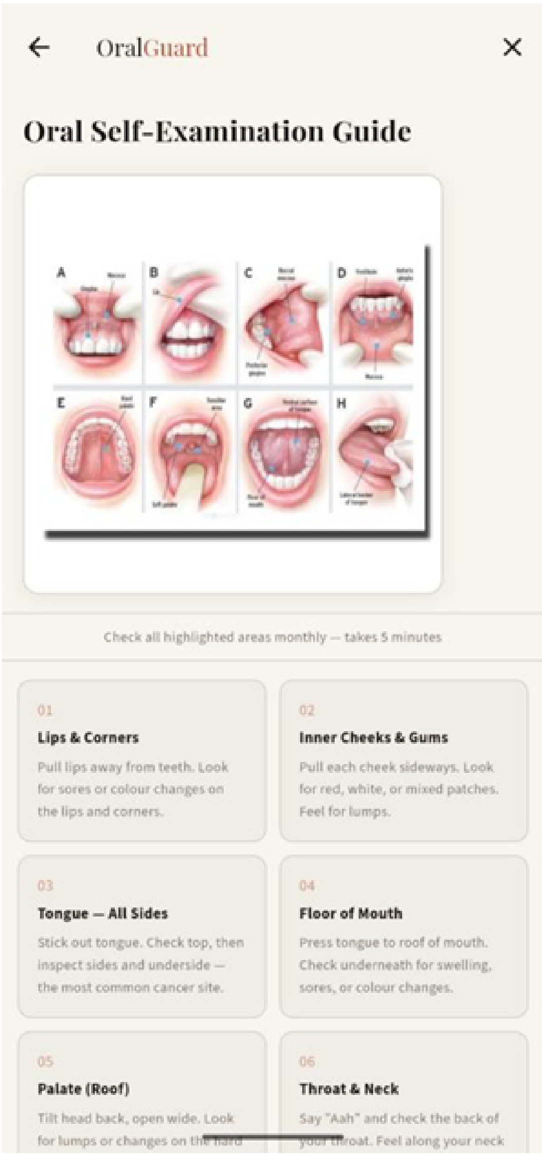

Fig. 6b. Oral self-examination guide (mobile)

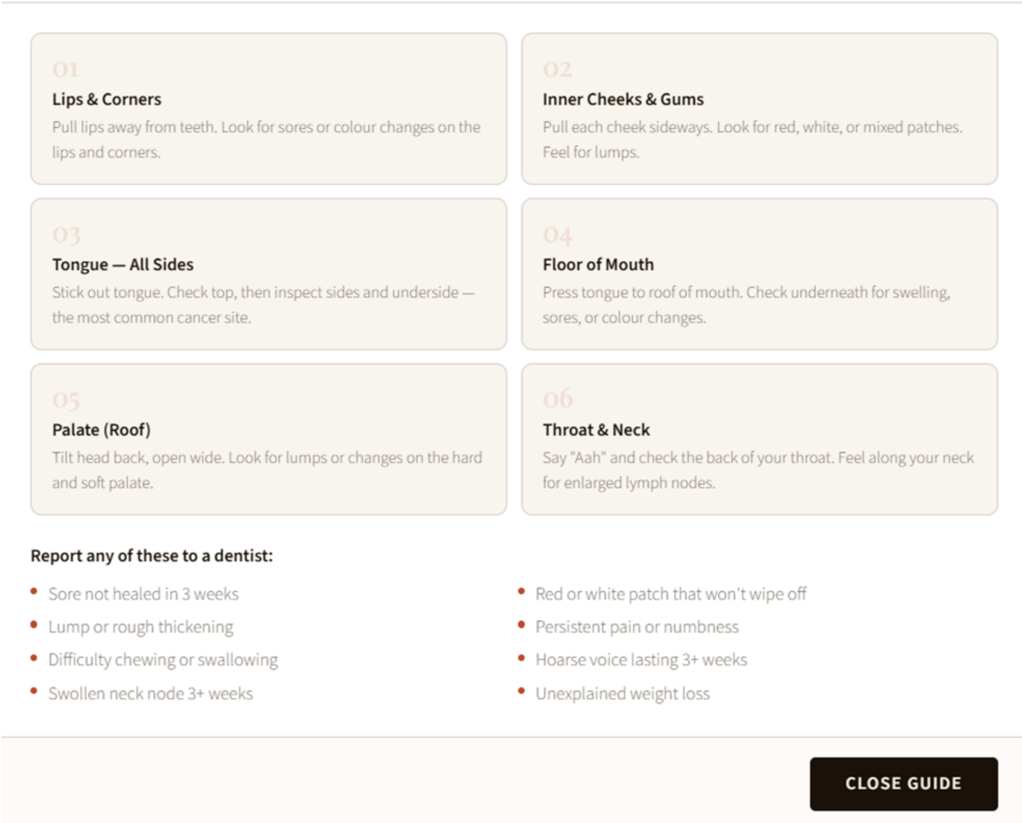

Fig. 7. Six-step self-examination protocol with symptom reporting guidance

Computational performance on the Hugging Face Spaces deployment was acceptable for a research prototype, with typical end-to-end response times of two to four seconds per query including network latency. This is not yet at a level appropriate for production healthcare deployment but is sufficient for the investigational and awareness context in which the platform currently operates.

The risk questionnaire is difficult to evaluate rigorously without outcome data, but the variable selection and weighting reflect well-established epidemiological associations. Tobacco use, betel nut consumption, and alcohol use are among the most consistently supported modifiable risk factors for oral squamous cell carcinoma. Incorporating HPV exposure, UV exposure for labial carcinoma, and family history captures additional dimensions of risk that tobacco and alcohol alone would miss.

4.1 Limitations

Several limitations of the current platform require explicit acknowledgment. The most significant is the scale and diversity of the underlying dataset. The Oral Cancer Image Dataset from Kaggle [18], while appropriate for a proof-of-concept implementation, contains a relatively small number of images concentrated in a limited range of clinical presentations and imaging conditions. CBIR performance in medical imaging tends to improve substantially with dataset size and diversity; a reference index built from thousands of clinically verified cases across multiple institutions and geographic populations would likely yield meaningfully better and more generalizable retrieval quality than the current implementation.

A second major limitation is the complete absence of formal user validation. The platform has not been evaluated in any structured study involving either clinical professionals or the target community user population. There is therefore no empirical evidence that the retrieval outputs are interpreted in the way the design intends, that the risk questionnaire responses are accurate, that the self-examination guidance leads to images of adequate quality, or that use of the platform actually influences health-seeking behavior in the intended direction. These are not minor gaps: they represent the core empirical questions that any awareness-focused health platform must answer before its utility can be meaningfully claimed. The system should be understood as an engineering and design prototype, not a validated health intervention.

Additional technical limitations include the use of a single backbone architecture without comparative evaluation of alternatives, the absence of formal precision-recall or retrieval accuracy metrics on held-out data, and the lack of adversarial testing for edge cases such as non-oral images or heavily distorted inputs. These issues are noted as priorities for future work.

5 Conclusion

OralGuard is a proof-of-concept platform for oral cancer risk awareness, combining content-based image retrieval, behavioral risk assessment, and guided self-examination into a smartphone-accessible tool. The platform is built around MobileNetV2 feature extraction and cosine similarity-based retrieval from an annotated reference index, deployed via FastAPI on Hugging Face Spaces and accessible through both a Flutter mobile application and a Gradio web interface.

The core design argument is that visual comparison against confirmed clinical cases is more interpretable and more likely to support appropriate health behavior in non-clinical users than classification probability outputs. Presenting parallel benign and malignant reference pools, rather than a single prediction, preserves comparative context and is intended to encourage professional consultation rather than self-diagnosis.

The platform has substantial limitations in its current form, primarily the limited scale of the training dataset and the absence of formal user validation. Before OralGuard or any similar system could be responsibly deployed in community health contexts, these gaps would need to be addressed through expansion of the reference dataset with multicenter clinical data, and through rigorous user studies assessing both interpretive accuracy and behavioral outcomes. Integration with telemedicine workflows, allowing retrieval outputs to be shared directly with remote specialists, represents a natural extension that could substantially increase the platform's practical utility.

The work contributes primarily at the level of architecture and concept: demonstrating the feasibility of integrating CBIR into a mobile-accessible oral cancer awareness platform, and establishing a design framework that could be built upon as better-annotated datasets and formal validation evidence become available.

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